4.3 Boundary behavior

In what follows we shall consider a **polygonal region** P, namely a bounded, simply connected open set whose boundary is a polygonal line $\mathfrak{p}$. In this context, we always assume that the polygonal line is closed, and we sometimes refer to $\mathfrak{p}$ as a **polygon**.

To study conformal maps from the half-plane $\mathbb{H}$ to P, we consider first the conformal maps from the disc $\mathbb{D}$ to P, and their boundary behavior.

Theorem 4.2 *If $F : \mathbb{D} \to P$ is a conformal map, then F extends to a continuous bijection from the closure $\overline{\mathbb{D}}$ of the disc to the closure $\overline{P}$ of the polygonal region. In particular, F gives rise to a bijection from the boundary of the disc to the boundary polygon $\mathfrak{p}$.*

The main point consists in showing that if z_0 belongs to the unit circle, then $\lim_{z \to z_0} F(z)$ exists. To prove this, we need a preliminary result, which uses the fact that if $f : U \to f(U)$ is conformal, then

$$\text{Area}(f(U)) = \int\!\!\int_U |f'(z)|^2 \, dx \, dy.$$

This assertion follows from the definition, $\text{Area}(f(U)) = \int\!\int_{f(U)} dx\, dy$, and the fact that the determinant of the Jacobian in the change of variables $w = f(z)$ is simply $|f'(z)|^2$, an observation we made in equation (4), Section 2.2, Chapter 1.

Lemma 4.3 *For each $0 < r < 1/2$, let C_r denote the circle centered at z_0 of radius r. Suppose that for all sufficiently small r we are given two points z_r and z_r' in the unit disc that also lie on C_r. If we let $\rho(r) = |f(z_r) - f(z_r')|$, then there exists a sequence $\{r_n\}$ of radii that tends to zero, and such that $\lim_{n \to \infty} \rho(r_n) = 0$.*

Proof. If not, there exist $0 < c$ and $0 < R < 1/2$ such that $c \leq \rho(r)$ for all $0 < r \leq R$. Observe that

$$f(z_r) - f(z_r') = \int_\alpha f'(\zeta)\, d\zeta,$$

where the integral is taken over the arc α on C_r that joins z_r and z_r' in $\mathbb{D}$. If we parametrize this arc by $z_0 + re^{i\theta}$ with $\theta_1(r) \leq \theta \leq \theta_2(r)$, then

$$\rho(r) \leq \int_{\theta_1(r)}^{\theta_2(r)} |f'(z)| r \, d\theta.$$

We now apply the Cauchy-Schwarz inequality to see that

$$\rho(r) \le \left(\int_{\theta_1(r)}^{\theta_2(r)} |f'(z)|^2 r \, d\theta \right)^{1/2} \left(\int_{\theta_1(r)}^{\theta_2(r)} r \, d\theta \right)^{1/2}.$$

Squaring both sides and dividing by r yields

$$\frac{\rho(r)^2}{r} \le 2\pi \int_{\theta_1(r)}^{\theta_2(r)} |f'(z)|^2 r \, d\theta.$$

We may now integrate both sides from 0 to R, and since $c \le \rho(r)$ on that region we obtain

$$c^2 \int_0^R \frac{dr}{r} \le 2\pi \int_0^R \int_{\theta_1(r)}^{\theta_2(r)} |f'(z)|^2 r \, d\theta dr \le 2\pi \int\!\!\int_{\mathbb{D}} |f'(z)|^2 \, dxdy.$$

Now the left-hand side is infinite because $1/r$ is not integrable near the origin, and the right-hand side is bounded because the area of the polygonal region is bounded, so this yields the desired contradiction and concludes the proof of the lemma.

Lemma 4.4 *Let z_0 be a point on the unit circle. Then $F(z)$ tends to a limit as z approaches z_0 within the unit disc.*

Proof. If not, there are two sequences $\{z_1, z_2, \ldots\}$ and $\{z_1', z_2', \ldots\}$ in the unit disc that converge to z_0 and are so that $F(z_n)$ and $F(z_n')$ converge to two distinct points ζ and ζ' in the closure of P. Since F is conformal, the points ζ and ζ' must lie on the boundary $\mathfrak{p}$ of P. We may therefore choose two disjoint discs D and D' centered at ζ and ζ', respectively, that are at a distance $d > 0$ from each other. For all large n, $F(z_n) \in D$ and $F(z_n') \in D'$. Therefore, there exist two continuous curves[8] Λ and Λ' in $D \cap P$ and $D' \cap P$, respectively, with $F(z_n) \in \Lambda$ and $F(z_n') \in \Lambda'$ for all large n, and with the end-points of Λ and Λ' equal to ζ and ζ', respectively.

Define $\lambda = F^{-1}(\Lambda)$ and $\lambda' = F^{-1}(\Lambda')$. Then λ and λ' are two continuous curves in $\mathbb{D}$. Moreover, both λ and λ' contain infinitely many points in each sequence $\{z_n\}$ and $\{z_n'\}$. Recall that these sequences converge to z_0. By continuity, the circle C_r centered at z_0 and of radius r will intersect λ and λ' for all small r, say at some points $z_r \in \lambda$ and $z_r' \in \lambda'$. This

[8]By a continuous curve, we mean the image of a continuous (not necessarily piecewise-smooth) function from a closed interval $[a, b]$ to $\mathbb{C}$.

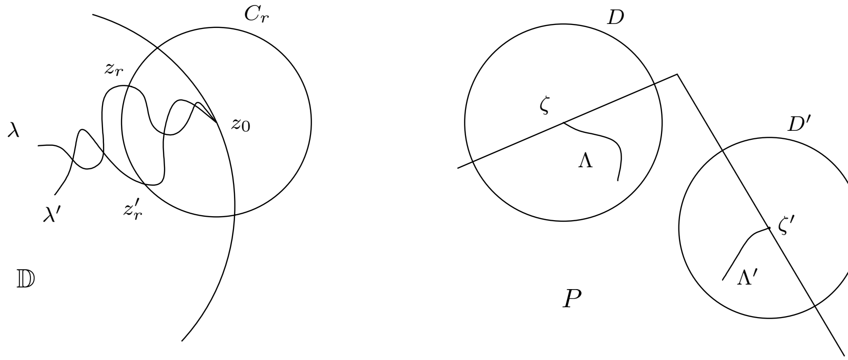

Figure 7. Illustration for the proof of Lemma 4.4

contradicts the previous lemma, because $|F(z_r) - F(z_r')| > d$. Therefore $F(z)$ converges to a limit on $\mathfrak{p}$ as z approaches z_0 from within the unit disc, and the proof is complete.

Lemma 4.5 *The conformal map F extends to a continuous function from the closure of the disc to the closure of the polygon.*

Proof. By the previous lemma, the limit

$$\lim_{z \to z_0} F(z)$$

exists, and we define $F(z_0)$ to be the value of this limit. There remains to prove that F is continuous on the closure of the unit disc. Given ϵ, there exists δ such that whenever $z \in \mathbb{D}$ and $|z - z_0| < \delta$, then $|F(z) - F(z_0)| < \epsilon$. Now if z belongs to the boundary of $\mathbb{D}$ and $|z - z_0| < \delta$, then we may choose w such that $|F(z) - F(w)| < \epsilon$ and $|w - z_0| < \delta$. Therefore

$$|F(z) - F(z_0)| \leq |F(z) - F(w)| + |F(w) - F(z_0)| < 2\epsilon,$$

and the lemma is established.

We may now complete the proof of the theorem. We have shown that F extends to a continuous function from $\overline{\mathbb{D}}$ to $\overline{P}$. The previous argument can be applied to the inverse G of F. Indeed, the key geometric property of the unit disc that we used was that if z_0 belongs to the boundary of $\mathbb{D}$, and C is any small circle centered at z_0, then $C \cap \mathbb{D}$ consists of an arc. Clearly, this property also holds at every boundary point of the polygonal region P. Therefore, G also extends to a continuous function from $\overline{P}$ to $\overline{\mathbb{D}}$. It suffices to now prove that the extensions of F and G are inverses of each other. If $z \in \partial\mathbb{D}$ and $\{z_k\}$ is a sequence in the disc that converges to z, then $G(F(z_k)) = z_k$, so after taking the limit and using the fact that F is continuous, we conclude that $G(F(z)) = z$ for all $z \in \overline{\mathbb{D}}$. Similarly, $F(G(w)) = w$ for all $w \in \overline{P}$, and the theorem is proved.

The circle of ideas used in this proof can be used to prove more general theorems on the boundary continuity of conformal maps. See Exercise 18 and Problem 6 below.

4.4 The mapping formula

Suppose P is a polygonal region bounded by a polygon $\mathfrak{p}$ whose vertices are ordered consecutively $a_1, a_2, \ldots, a_n$, and with $n \geq 3$. We denote by $\pi\alpha_k$ the interior angle at a_k, and define the exterior angle $\pi\beta_k$ by $\alpha_k + \beta_k = 1$. A simple geometric argument provides $\sum_{k=1}^{n} \beta_k = 2$.

We shall consider conformal mappings of the half-plane $\mathbb{H}$ to P, and make use of the results of the previous section regarding conformal maps from the disc $\mathbb{D}$ to P. The standard correspondences $w = (i - z)/(i + z)$, $z = i(1 - w)/(1 + w)$ allows us to go back and forth between $z \in \mathbb{H}$ and $w \in \mathbb{D}$. Notice that the boundary point $w = -1$ of the circle corresponds to the point at infinity on the line, and so the conformal map of $\mathbb{H}$ to $\mathbb{D}$ extends to a continuous bijection of the boundary of $\mathbb{H}$, which for the purpose of this discussion includes the point at infinity.

Let F be a conformal map from $\mathbb{H}$ to P. (Its existence is guaranteed by the Riemann mapping theorem and the previous discussion.) We assume first that none of the vertices of $\mathfrak{p}$ correspond to the point at infinity. Therefore, there are real numbers $A_1, A_2, \ldots, A_n$ so that $F(A_k) = a_k$ for all k. Since F is continuous and injective, and the vertices are numbered consecutively, we may conclude that the A_k's are in either increasing or decreasing order. After relabeling the vertices a_k and the points A_k, we may assume that $A_1 < A_2 < \cdots < A_n$. These points divide the real line into $n - 1$ segments $[A_k, A_{k+1}]$, $1 \leq k \leq n - 1$, and the segment that consists of the join of the two half-segments $(-\infty, A_1] \cup [A_n, \infty)$. These are mapped bijectively onto the corresponding sides of the polygon, that is, the segments $[a_k, a_{k+1}]$, $1 \leq k \leq n - 1$, and $[a_n, a_1]$ (see Figure 8).

Theorem 4.6 *There exist complex numbers c_1 and c_2 so that the conformal map F of $\mathbb{H}$ to P is given by*

$$F(z) = c_1 S(z) + c_2$$

where S is the Schwarz-Christoffel integral introduced in Section 4.2.